Lab – 04

Implement SQL Functions: Conversion Functions, and Date Functions.

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);

INSERT INTO STUDENT VALUES
(101, 'Aarav Sharma', '10th', 485, 25000.00, '2008-05-14', '2022-04-10'),
(102, 'Ananya Verma', '10th', 462, 25000.00,'2008-09-22', '2022-04-12'),
(103, 'Rohan Mehta', '9th', 390, 18500.50, '2009-01-30', '2023-04-05'),
(104, 'Priya Singh', '11th', 495, 30000.00, '2007-11-08', '2021-06-15'),
(105, 'Kabir Nair', '9th', 415, 20000.00, '2009-07-19', '2023-04-18');
SELECT * FROM Student;
```

SQL-functions.sql

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);

INSERT INTO STUDENT VALUES
(101, 'Aarav Sharma', '10th', 485, 25000.00, '2008-05-14', '2022-04-10'),
(102, 'Ananya Verma', '10th', 462, 25000.00, '2008-09-22', '2022-04-12'),
(103, 'Rohan Mehta', '9th', 390, 18500.50, '2009-01-30', '2023-04-05'),
(104, 'Priya Singh', '11th', 495, 30000.00, '2007-11-08', '2021-06-15'),
(105, 'Kabir Nair', '9th', 415, 20000.00, '2009-07-19', '2023-04-18');
SELECT * FROM Student;
```

Output:

Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
101	Aarav Sharma	10th	485	25000.00	2008-05-14	2022-04-10
102	Ananya Verma	10th	462	25000.00	2008-09-22	2022-04-12
103	Rohan Mehta	9th	390	18500.50	2009-01-30	2023-04-05
104	Priya Singh	11th	495	30000.00	2007-11-08	2021-06-15
105	Kabir Nair	9th	415	20000.00	2009-07-19	2023-04-18

```
/* Common String Functions */
SELECT Student_Name,
```

```
   UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

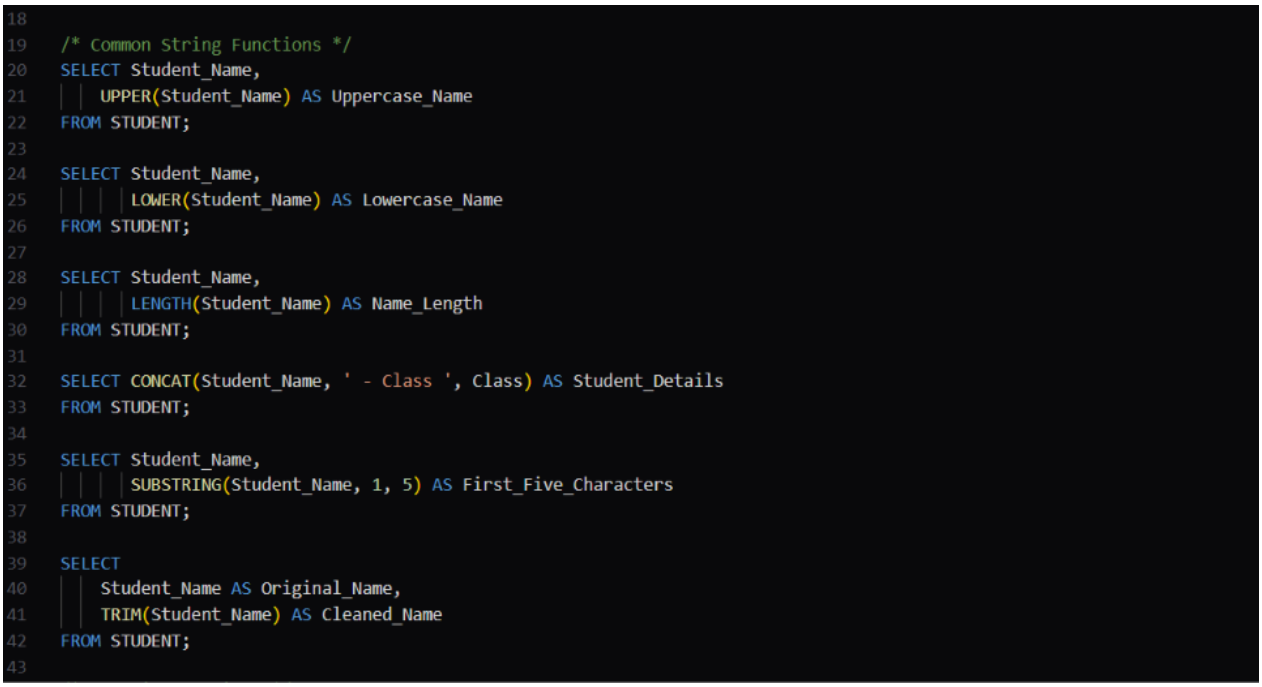
SELECT Student_Name,
      LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
      LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
      SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT
   Student_Name AS Original_Name,
   TRIM(Student_Name) AS Cleaned_Name
FROM STUDENT;
```



```sql

/* Common String Functions */
SELECT Student_Name,
    UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
        LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
        LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
        SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT
    Student_Name AS Original_Name,
    TRIM(Student_Name) AS Cleaned_Name
FROM STUDENT;

```

Student_Name	Uppercase_Name
Aarav Sharma	AARAV SHARMA
Ananya Verma	ANANYA VERMA
Rohan Mehta	ROHAN MEHTA
Priya Singh	PRIYA SINGH
Kabir Nair	KABIR NAIR

Student_Name	Lowercase_Name
Aarav Sharma	aarav sharma
Ananya Verma	ananya verma
Rohan Mehta	rohan mehta
Priya Singh	priya singh
Kabir Nair	kabir nair

Student_Name	Name_Length
Aarav Sharma	12
Ananya Verma	12
Rohan Mehta	11
Priya Singh	11
Kabir Nair	10

Student_Details
Aarav Sharma - Class 10th
Ananya Verma - Class 10th
Rohan Mehta - Class 9th
Priya Singh - Class 11th
Kabir Nair - Class 9th

Student_Name	First_Five_Characters
Aarav Sharma	Aarav
Ananya Verma	Anany
Rohan Mehta	Rohan
Priya Singh	Priya
Kabir Nair	Kabir

Original_Name	Cleaned_Name
Aarav Sharma	Aarav Sharma
Ananya Verma	Ananya Verma
Rohan Mehta	Rohan Mehta
Priya Singh	Priya Singh
Kabir Nair	Kabir Nair

```
/* Numeric Functions */
SELECT Student_Name,
    Marks,
    ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
    Marks / 10.0 AS Marks_Value,
    CEIL(Marks / 10.0) AS Rounded_Up
```

```
FROM STUDENT;

SELECT Student_Name,
    Marks / 10.0 AS Marks_Value,
    FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
    Marks,
    MOD(Marks, 10) AS Remainder
FROM STUDENT;
SELECT Student_Name,
    Marks,
    ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;
```

```
/* Numeric Functions */
SELECT Student_Name,
        Marks,
        ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

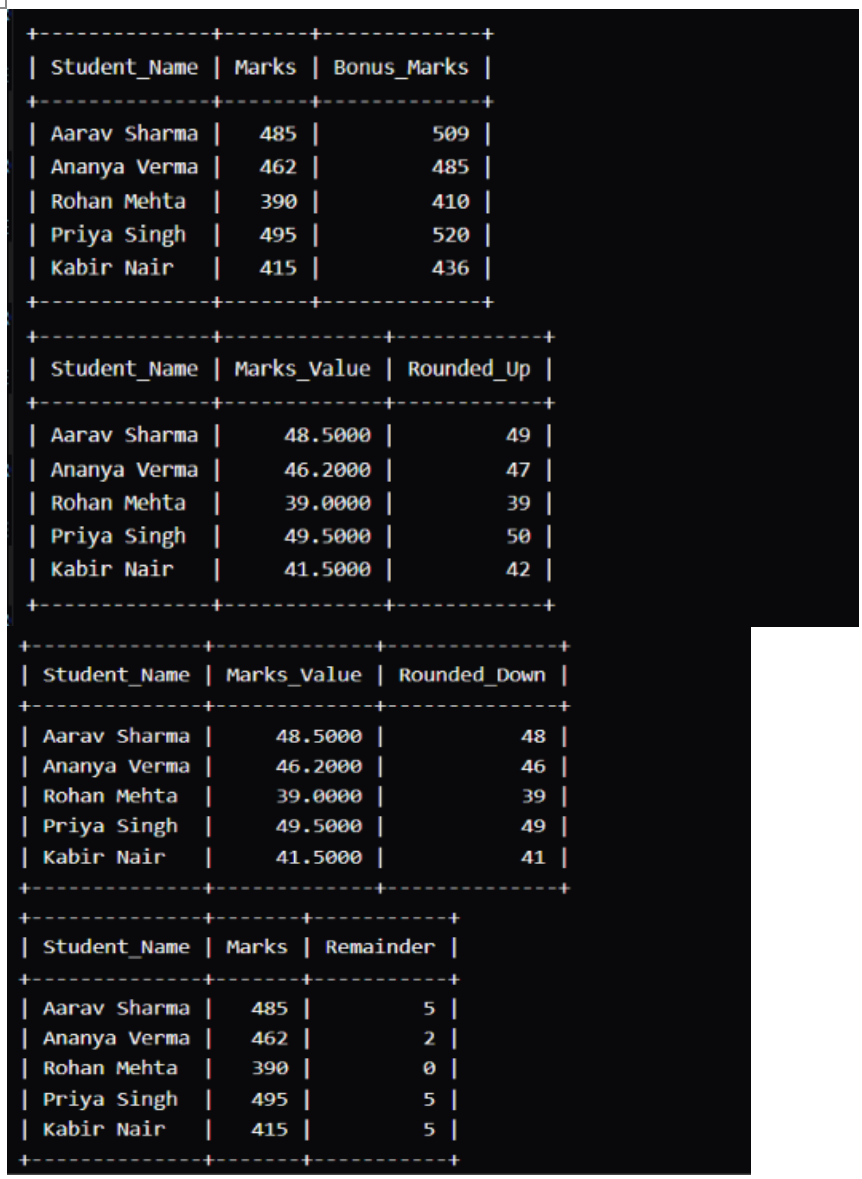
SELECT Student_Name,
        Marks / 10.0 AS Marks_Value,
        CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
        Marks / 10.0 AS Marks_Value,
        FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
        Marks,
        MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
        Marks,
        ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;
```

```
+--------------+-------+--------------------+
| Student_Name | Marks | Difference_From_80 |
+--------------+-------+--------------------+
| Aarav Sharma |   485 |                405 |
| Ananya Verma |   462 |                382 |
| Rohan Mehta  |   390 |                310 |
| Priya Singh  |   495 |                415 |
| Kabir Nair   |   415 |                335 |
+--------------+-------+--------------------+
```



Student_Name	Marks	Bonus_Marks
Aarav Sharma	485	509
Ananya Verma	462	485
Rohan Mehta	390	410
Priya Singh	495	520
Kabir Nair	415	436

Student_Name	Marks_Value	Rounded_Up
Aarav Sharma	48.5000	49
Ananya Verma	46.2000	47
Rohan Mehta	39.0000	39
Priya Singh	49.5000	50
Kabir Nair	41.5000	42

Student_Name	Marks_Value	Rounded_Down
Aarav Sharma	48.5000	48
Ananya Verma	46.2000	46
Rohan Mehta	39.0000	39
Priya Singh	49.5000	49
Kabir Nair	41.5000	41

Student_Name	Marks	Remainder
Aarav Sharma	485	5
Ananya Verma	462	2
Rohan Mehta	390	0
Priya Singh	495	5
Kabir Nair	415	5

```
/* Date Functions */
SELECT Student_Name,
    Date_of_Birth,
    YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today;
```

```sql
/* Date Functions */
SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today;
```

Student_Name	Date_of_Birth	Birth_Day
Aarav Sharma	2008-05-14	14
Ananya Verma	2008-09-22	22
Rohan Mehta	2009-01-30	30
Priya Singh	2007-11-08	8
Kabir Nair	2009-07-19	19

Student_Name	Date_of_Birth	Age
Aarav Sharma	2008-05-14	18
Ananya Verma	2008-09-22	17
Rohan Mehta	2009-01-30	17
Priya Singh	2007-11-08	18
Kabir Nair	2009-07-19	17

Student_Name	Date_of_Birth	Birth_Year
Aarav Sharma	2008-05-14	2008
Ananya Verma	2008-09-22	2008
Rohan Mehta	2009-01-30	2009
Priya Singh	2007-11-08	2007
Kabir Nair	2009-07-19	2009

Student_Name	Date_of_Birth	Birth_Month
Aarav Sharma	2008-05-14	5
Ananya Verma	2008-09-22	9
Rohan Mehta	2009-01-30	1
Priya Singh	2007-11-08	11
Kabir Nair	2009-07-19	7

```
/* AGGREGATE FUNCTIONS */
SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;
```

